

Mohammad Salim Hosen

Senior Full-Stack & AI Systems Engineer

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PROFESSIONAL SUMMARY

Results-driven Senior Full-Stack & AI Systems Engineer with **6+ years of experience** architecting resilient web platforms, high-throughput distributed scraping engines, and production AI integrations. Deep expertise in backend system design with **PHP (Laravel, CodeIgniter), Python (FastAPI, Scrapy), Node.js**, and columnar big-data analytics in **ClickHouse**. Proven track record deploying autonomous extraction agents, vector search pipelines, and RESTful APIs powering enterprise SaaS and marketplace platforms. Adept at cutting database latency (<120ms), driving automated CI/CD releases on AWS, and delivering high-impact solutions across cross-functional international remote teams.

TECHNICAL COMPETENCIES

AI & Intelligent Systems:	LLM APIs (OpenAI, Claude, Gemini), Vector Search & Embeddings, RAG Architectures, Autonomous Scraping Agents, Semantic Search & Caching, Prompt Architecture
Backend & Distributed Systems:	PHP (Laravel, CodeIgniter 4), Python (FastAPI, Scrapy, BeautifulSoup), Node.js, RESTful API Design, Microservices, Asynchronous Queue Workers, System Design
Databases & Columnar Big Data:	ClickHouse (High-Volume Columnar Analytics), PostgreSQL, MySQL, Redis (Caching & Queues), MongoDB
Frontend Engineering:	React.js, Next.js, Vue.js, TypeScript, JavaScript (ES6+), Modern CSS3, Tailwind CSS, HTML5
Cloud, DevOps & Tools:	AWS (EC2, S3, RDS), Docker & Containers, GitHub Actions (CI/CD), Nginx, Linux, Git, Agile/Scrum

WORK EXPERIENCE

Byteshake Ltd.	Dhaka, Bangladesh • Hybrid / Remote
Senior Software Engineer • Backend Architecture & AI Systems	Mar 2025 – Present
- Spearheaded backend architecture for Tenderbase, an enterprise platform aggregating over 500,000+ tenders across global public and commercial procurement portals. - Architected an AI-powered semantic search and ranking engine combining vector embeddings with ClickHouse indexing, boosting user discovery relevance by 40%. - Engineered distributed Python scraping engines with automated proxy rotation, rate-limiting, and resilient LLM fallback parsers for dynamic, unstructured supplier sites. - Optimized high-frequency analytical queries and ingestion pipelines in ClickHouse, reducing complex multi-million row reporting latency from 3.2s to under 120ms. - Constructed low-latency RESTful APIs using Laravel and CodeIgniter 4 with Redis caching and asynchronous queues, maintaining consistent sub-80ms response times. - Administered cloud deployment infrastructure on AWS (EC2, RDS, S3) with automated GitHub Actions CI/CD, ensuring zero-downtime production releases.	
CureTech Ltd.	Dhaka, Bangladesh • Full-time
Senior Software Engineer • Full-Stack & Automation	Jan 2022 – Mar 2025
- Directed full-stack architecture for enterprise CRM and multi-warehouse inventory systems handling thousands of critical daily transactions. - Developed custom Python web scrapers and data extraction pipelines for automated competitor price intelligence and catalog synchronization. - Architected secure, high-throughput REST APIs powering mobile delivery management applications with JWT authentication and real-time dispatch tracking. - Built modern, responsive frontends in React.js and Next.js integrated with Laravel backends, cutting initial page load times by 35%. - Maintained 99.9% application uptime across production systems using Docker containerization and GitHub Actions automated test suites.	

- Developed a high-volume multivendor e-commerce marketplace from scratch using Laravel, Vue.js, and MySQL, featuring dynamic commission calculations and automated payouts.
- Built an automated company data scraping engine in Python to extract, normalize, and enrich commercial records for sales intelligence pipelines.
- Architected relational database schemas and indexed queries handling concurrent shopping carts and checkout transactions.
- Collaborated in an agile engineering team practicing code reviews, sprint planning, and test-driven feature development.

KEY PROJECTS & SYSTEM ARCHITECTURES

Tenderbase – Python, ClickHouse, Codelgniter 4, AI Semantic Search, AWS [\[tenderbase.co.uk\]](https://tenderbase.co.uk)

High-scale procurement intelligence platform indexing 500k+ global tenders. Engineered distributed Python scrapers, AI-driven semantic search, and columnar query optimization in ClickHouse for sub-second analytical reporting.

TapToConnect – Laravel, React.js, REST API, NFC, Redis, MySQL [\[my.taptoconnect.app\]](https://my.taptoconnect.app)

End-to-end SaaS platform facilitating digital identity sharing via NFC technology, featuring advanced admin controls, real-time analytics dashboard, and automated customer contact exchange workflows.

Revuity – React.js, Node.js, Supabase, PostgreSQL, Tailwind CSS [\[revuity.com\]](https://revuity.com)

Customer feedback collection and reputation management platform with real-time survey delivery, automated follow-up notification triggers, and customer satisfaction metrics dashboards.

PickGo – Laravel, Vue.js, Real-time WebSockets, MySQL [\[Proprietary System\]](#)

Proprietary Restaurant Order Management SaaS featuring real-time order processing, complex dynamic inventory algorithms, and deep third-party POS integrations.

Asoug.com – Laravel, Vue.js, REST APIs, MySQL, Payment Gateways [\[asoug.com\]](https://asoug.com)

Full-featured B2B and multi-vendor e-commerce platform handling multi-category inventory, individual vendor dashboards, secure payment gateways, and automated order fulfillment pipelines.

EDUCATION

Daffodil International University • Dhaka, Bangladesh

2016 – 2020

Bachelor of Science in Computer Science & Engineering (B.Sc. in CSE)

Focus: Algorithms, Data Structures, Distributed Systems, Database Architecture.

Active competitive programming participant (ACM-ICPC format contests); Member of DIU Computer Programming Club (CPC).

LANGUAGES & REMOTE WORK READINESS

Languages: English (Full Professional Working Proficiency), Bengali (Native)

Remote Collaboration: Asynchronous Autonomy, Flexible 4–5 hours daily overlap with US (EST/PST), UK (GMT), and European (CET) timezones.